

Measurement Review — Volume, Conversions & Angles Answer Key

Grade 5

Quick Answers

- | | |
|------------------------|---|
| 1. 300 cm | 8. 120 degrees |
| 2. 24 cm ³ | 9. 135 min |
| 3. 65 degrees | 10. 6 cm |
| 4. 2500 g | 11. 8 |
| 5. 55 degrees | 12. (a) volume in cubic centimeters = 30000 cm ³ , |
| 6. 100 cm ³ | (b) liters = 30 L, — |
| 7. 1750 m | |
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What is verified

11 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 5

4.MD – *problems 3, 5, 8, 9*

5.MD – *problems 1, 2, 4, 6, 7, 10, 11, 12*

Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

4. *If they got 250: used 100 instead of 1000 grams per kilogram*
10. *If they got 18: divided the volume by only one base dimension*
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1. Convert 3 m to centimeters

Solution: Each meter is 100 centimeters, so 3 meters is 3×100 .

300 cm

2. Tissue box $4 \times 2 \times 3$ cm

Solution: $V = \ell \times w \times h = 4 \times 2 \times 3$. The base layer holds $4 \times 2 = 8$ centimeter cubes and there are 3 layers.

24 cm ³

3. Straight-line pair with 115°

Solution: Angles on a straight line total 180° , so the other angle is $180 - 115 = 65$.

65°

4. Rice: 2.5 kg in grams

Solution: Each kilogram is 1000 grams: $2.5 \times 1000 = 2500$.

2500 g

5. Right angle split, one part 35°

Solution: A right angle totals 90° , so the other part is $90 - 35 = 55$.

55°

6. Step stool: two boxes, 5×4 base, heights 3 and 2

Solution: Both boxes share the 5×4 base, so stack the heights first: $3 + 2 = 5$ cm tall in all. Then $V = 5 \times 4 \times 5 = 100$. (Or add the two volumes: $60 + 40$.)

100 cm^3

7. Fun run: 3 km, 1250 m already run

Solution: Convert first so both distances use the same unit: $3 \text{ km} = 3 \times 1000 = 3000$ m. Then subtract: $3000 - 1250 = 1750$.

1750 m

8. Minute hand: 360° in 60 minutes; 20 minutes

Solution: Each minute the hand sweeps $360 \div 60 = 6$ degrees. In 20 minutes: $6 \times 20 = 120$.

120°

9. Movie: 2 h 15 min in minutes

Solution: Each hour is 60 minutes: $2 \times 60 = 120$, plus the extra 15 gives 135.

135 min

10. Box: volume 72 cm^3 , base 4×3 cm — height?

Solution: The base area is $4 \times 3 = 12$, so each layer of cubes holds 12. The number of layers is $72 \div 12 = 6$.

6 cm

11. Jug: 2 L, cups of 250 mL

Solution: Convert the jug to milliliters: $2 \times 1000 = 2000$ mL. Then $2000 \div 250 = 8$. The jug fills
cups.

8

12. Challenge: fish tank $50 \times 20 \times 30$ cm

Solution: (a) $V = 50 \times 20 \times 30$: base $50 \times 20 = 1000$, times 30 layers.

30000 cm ³

(b) $30000 \div 1000 = 30$.

30 L

(c) Every 1000 cubic centimeters makes one liter, so dividing the cubic-centimeter count by 1000 counts how many liter-sized groups fit — that is exactly what converting to liters means. A correct response must connect the division to regrouping 1000 cm³ per liter.